1. Introduction to Ligand–Protein Interactions

- **Definition**: Ligand–protein interaction refers to the binding of a small molecule (ligand) to a specific site on a protein, often leading to a biological response.
- **Types of interactions**:
    - Hydrogen bonding
    - Hydrophobic interactions
    - Ionic interactions
    - van der Waals forces
    - π–π stacking
- **Importance**:
    - Basis of drug action.
    - Critical in enzyme-substrate binding, receptor signaling, and protein regulation.
    - Understanding these helps in rational drug design.

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2. Concept of Molecular Docking Studies

- **Definition**: Molecular docking predicts the preferred orientation of a ligand when bound to a protein, estimating the strength (binding affinity) of the interaction.
- **Objective**:
    1. Predict binding poses (conformations) and binding energies.
    2. Rank compounds based on docking scores.
- **Steps**:
    1. Receptor and ligand preparation.

2. Defining the binding site (grid).
3. Docking using algorithms (e.g., genetic algorithms, simulated annealing).
4. Scoring and ranking based on binding energy.

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3. Introduction to Small Molecule Tools

- **Definition**: Software used for designing, editing, and optimizing small molecule ligands.
- **Examples**:
    - **ChemSketch**: Draw and export 2D/3D ligand structures.
    - **Avogadro**: Build and optimize small molecules.
    - **Open Babel**: Convert between chemical formats.
    - **MarvinSketch**: Chemoinformatics and property prediction.
- **Purpose**: Prepare ligands for docking, including energy minimization and format conversion.

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4. Tools for Molecular Docking

- **AutoDock**: Widely used, free tool that uses Lamarckian genetic algorithms.
- **AutoDock Vina**: Faster, more accurate than AutoDock, simpler interface.
- **PyRx**: GUI for AutoDock/Vina with support for batch docking.
- **Schrödinger Glide**: Commercial tool, highly accurate.
- **SwissDock**: Web-based, AutoDock-based.
- **DockThor, GOLD, FlexX, MOE**: Other options with varying features.
- **VMD, Chimera, PyMOL**: Visualization tools used alongside docking tools.

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5. Installation of AutoDock

- **Prerequisites**:
    1. Python 2.7 (for AutoDockTools).
    2. MGLTools (for GUI).
- **Steps**:
    1. Download from AutoDock official site.
    2. Install MGLTools (includes AutoDockTools for file preparation).
    3. Set environment variables (on Windows or Linux).
    4. Install AutoDock binaries.
    5. Verify installation using command line or GUI.

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6. How to Prepare Ligand

- **Steps**:
    1. Draw the structure using ChemSketch/MarvinSketch.
    2. Convert to 3D and perform geometry optimization (Avogadro).
    3. Save as `.mol2` or `.pdb`.
    4. Use AutoDockTools:
        - Add hydrogens.
        - Compute Gasteiger charges.
        - Define rotatable bonds.

        - Save in `.pdbqt` format.

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7. How to Prepare Protein Files

- **Steps**:
    1. Download PDB file from RCSB PDB.
    2. Remove non-standard residues, water molecules, and unwanted chains.
    3. Add missing hydrogens.
    4. Assign Kollman charges.
    5. Save as `.pdbqt` using AutoDockTools.
- **Important**: The protein should be in rigid form unless flexible docking is used.

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8. Importance of Grid and Dock Files

- **Grid File (`.gpf`)**:
    - Defines the 3D region (active site) for docking.
    - Calculates grid maps for each atom type (pre-processing step).
- **Docking File (`.dpf`)**:
    - Contains docking parameters like search algorithm, number of runs, etc.
- **Why important**:
    - Without accurate grid and docking parameters, docking may yield unrealistic or non-specific results.

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9. Preparation of Grid Parameters

- **Steps**:
    1. Open AutoDockTools.
    2. Load receptor and ligand.
    3. Set grid box:
        - Center on active site.
        - Adjust box size (grid points) and spacing (default 0.375 Å).
    4. Set atom types and receptor file.
    5. Save as `.gpf`.
- **Tip**: Use known ligand binding site coordinates (from co-crystallized ligands).

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10. Preparation of Dock Parameters

- **Steps**:
    1. Select docking algorithm (e.g., Lamarckian Genetic Algorithm).
    2. Set:
        - Number of runs (e.g., 10–100).
        - Energy evaluations (e.g., 2.5 million).
        - Population size and generations.
    3. Link receptor and ligand `.pdbqt` files.
    4. Save `.dpf`.
- **Note**: Proper parameter tuning improves reproducibility and accuracy.

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11. Case Studies

Case Study 1: Docking an Antiviral Drug with SARS-CoV-2 Mpro

- **Objective**: Identify potential inhibitors from FDA-approved drugs.
- **Process**:
  - Target: Main protease (Mpro) of SARS-CoV-2.
  - Ligands: Remdesivir, Lopinavir, etc.
  - Tools: AutoDock Vina, PyRx.
  - Outcome: Binding affinities, interaction residues.

Case Study 2: Inhibitor Design for Acetylcholinesterase

- **Objective**: Develop neuroprotective inhibitors for Alzheimer's.
- **Steps**:
  - Protein: Acetylcholinesterase (PDB).
  - Ligands: Natural products library.
  - Result: Identified top 5 hits with H-bonds at the active site.

Case Study 3: Virtual Screening of Phytochemicals

- **Goal**: Screen compounds from medicinal plants against cancer targets.
- **Workflow**:
  - Prepare ligand library.
  - Dock with cancer-related enzymes.
  - Analyze binding pose and energy.

- **Impact**: Leads for in vitro validation.